The Cosmological Lithium Problem and Recognition Science: φ-Modulated BBN Reaction Rates

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Abstract

We analyze the cosmological lithium problem within Recognition Science (RS). Standard Big Bang Nucleosynthesis (BBN) predicts a primordial ^{7}Li abundance of $^7\mathrm{Li/H} \approx (4.7 \pm 0.7) \times 10^{-10}$, while observations of metal-poor halo stars (the Spite plateau) give $^7\mathrm{Li/H} \approx (1.6 \pm 0.3) \times 10^{-10}$—a factor of ~ 3 discrepancy [1, 2]. RS predicts a φ-modulated suppression of the key nuclear reaction rate $^3\mathrm{He}(\alpha,\gamma)^7\mathrm{Be}$ via the 8-tick recognition cycle. The per-cycle fractional modulation is $\delta_\varphi = \ln\varphi/2^D \approx 0.060$ (6%), derived from the RS φ-ladder and 8-tick period. The required factor of ~ 3 suppression is obtained after $N \approx 18$ effective recognition-cycle epochs; we show that $N \approx 18$ is order-of-magnitude consistent with the number of φ-rung transitions during the ^{7}Be production window, and note that a first-principles derivation of N from the RS Hamiltonian remains to be completed. The ^{4}He abundance and deuterium ratio are insensitive to this modulation.

1 Recognition Science Framework

The φ-modulation mechanism applied to BBN lithium production derives from the unique cost functional fixed by the RS axioms.

Definition 1.1 (RS Cost Functional)**.** For $x > 0$: $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Axiom A1 (Normalization): $J(1) = 0$.
Axiom A2 (Recognition Composition Law, RCL): $J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y)$ for all $x, y > 0$.
Axiom A3 (Calibration): $J''_{\log}(0) = 1$, where $J_{\log}(t) := J(e^t)$.

Theorem 1.2 (Cost Uniqueness)**.** *The continuous solution to* (A1)–(A3) *is unique:* $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Proof sketch. Set $H(t) = J_{\log}(t) + 1$. Axiom A2 transforms into the d'Alembert equation $H(s + t) + H(s - t) = 2H(s)H(t)$. By the Aczél classification theorem [7], the unique continuous solution with $H(0) = 1$ and $H''(0) = 1$ is $H(t) = \cosh t$, giving $J(x) = \frac{1}{2}(x + x^{-1}) - 1$. □

The φ-ladder spacing $\ln\varphi$ distributes over the 8-tick recognition epoch ($2^D = 8$, $D = 3$ forced by $2^D = 8$) giving the per-epoch modulation $\delta_\varphi = \ln\varphi/2^D \approx 0.060$. This is the same universal factor that appears in the gallium anomaly resolution; it is a structural prediction of the φ-ladder and 8-tick period, not a free parameter.

2 Introduction

Big Bang Nucleosynthesis is one of the pillars of the hot Big Bang model. The predicted abundances of ^{4}He, D, ^{3}He, and ^{7}Li depend sensitively on the baryon-to-photon ratio η, independently determined by Planck as $\eta = 6.1 \times 10^{-10}$. For ^{4}He and D, BBN predictions agree with observations to better than 1%. For ^{7}Li, they disagree by a factor of ~ 3.

The *cosmological lithium problem* [3, 1] is this factor-of-3 discrepancy:

$$\left(\frac{^7\mathrm{Li}}{\mathrm{H}}\right)_{\mathrm{BBN}} \approx (4.7 \pm 0.7) \times 10^{-10}, \tag{1}$$

$$\left(\frac{^7\mathrm{Li}}{\mathrm{H}}\right)_{\mathrm{Spite}} \approx (1.6 \pm 0.3) \times 10^{-10}. \tag{2}$$

The Spite plateau is the remarkably constant ^{7}Li abundance observed in metal-poor halo stars ([Fe/H] < -1.5), believed to reflect the primordial value.

Proposed solutions include (i) stellar depletion by mixing below the convective zone, (ii) non-standard BBN (extra neutrino species, varying constants), and (iii) nuclear physics uncertainties in reaction rates. None fully resolves the problem without other tensions [2].

Recognition Science [6] (RS) proposes a fourth option: the 8-tick recognition cycle suppresses the $^3\mathrm{He}(\alpha,\gamma)^7\mathrm{Be}$ reaction rate during BBN by a derived fractional factor, accumulated over the ^{7}Be production window.

3 Standard BBN Lithium Production

Primordial ^{7}Li is produced primarily through:

$$^3\mathrm{He} + {}^4\mathrm{He} \to {}^7\mathrm{Be} + \gamma, \qquad {}^7\mathrm{Be} + e^- \to {}^7\mathrm{Li} + \nu_e. \tag{3}$$

The ^{7}Be production window opens at $T \sim 0.4$ MeV (approximately 100 s after the Big Bang) and closes at $T \sim 0.1$ MeV (~ 1000 s), as the temperature drops below the threshold for nuclear reactions. The ^{7}Li yield is directly proportional to the integrated $^3\mathrm{He}(\alpha,\gamma)^7\mathrm{Be}$ rate over this window.

4 The RS Modulation Mechanism

4.1 Per-Cycle Rate Modulation

Theorem 4.1 (Rate Modulation Factor)**.** *The 8-tick recognition cycle modulates nuclear reaction rates by a fractional factor:*

$$\delta_\varphi = \frac{\ln\varphi}{2^D} = \frac{\ln\varphi}{8} \approx 0.060 \quad (6\%), \tag{4}$$

where $D = 3$ is the number of spatial dimensions and $2^D = 8$ is the 8-tick period.

Proof. The φ-ladder spacing in log-space is $\ln\varphi \approx 0.481$. One 8-tick recognition epoch distributes this spacing uniformly across the 8 ticks. The per-tick fractional modulation is $\ln\varphi/8$. For a nuclear reaction whose cross section depends on the φ-phase of the recognition cycle, the per-epoch fractional modification to the rate is:

$$\delta_\varphi = \frac{\ln\varphi}{2^D} = \frac{\ln\varphi}{8} = \frac{0.4812}{8} \approx 0.0601.$$

The $^3\mathrm{He}(\alpha,\gamma)^7\mathrm{Be}$ reaction involves a transition between nuclear states of mass numbers $A = 3 + 4 = 7$ and $A = 7$, spanning one φ-rung in the nuclear binding energy hierarchy. The rung-crossing probability is modulated by δ_φ per epoch. □

Remark 4.2. The same 6% per-cycle modulation appears in the Gallium anomaly (the ν-nucleus cross section is suppressed by $57.7/74.0 \approx 0.780 \approx (1-\delta_\varphi)^4$ over four recognition epochs, corresponding to the $D + 1 = 4$ independent ledger dimensions traversed by a charged-current weak process). The modulation factor $\delta_\varphi = \ln\varphi/8$ is thus a universal RS prediction for electromagnetic and weak nuclear processes that cross φ-rungs.

4.2 Cumulative Suppression

Theorem 4.3 (Cumulative Suppression Factor)**.** *The cumulative suppression factor for ^{7}Be production over N effective recognition epochs is:*

$$F_{\mathrm{sup}}(N) = (1-\delta_\varphi)^N = \left(1 - \frac{\ln\varphi}{8}\right)^N. \tag{5}$$

For $N \approx 18$:

$$F_{\mathrm{sup}}(18) = (1-0.0601)^{18} \approx 0.328 \approx \frac{1}{3}. \tag{6}$$

Proof. At each effective recognition epoch, the ^{7}Be production rate is suppressed by factor $(1-\delta_\varphi)$. After N independent epochs, the rate is $(1-\delta_\varphi)^N$. Numerically:

$$(1-0.0601)^{18} = (0.9399)^{18}.$$

Computing $\ln(0.9399^{18}) = 18 \times \ln(0.9399) = 18 \times (-0.0619) = -1.114$, so $(0.9399)^{18} = e^{-1.114} \approx 0.328$. □

Remark 4.4 (Status of $N \approx 18$). The value $N \approx 18$ is derived by *inverting* the required suppression factor—it is the unique integer satisfying $F_{\mathrm{sup}}(N) \approx 1/3$ given $\delta_\varphi \approx 0.060$. Explicitly:

$$N = \frac{\ln(1/3)}{\ln(1-\delta_\varphi)} = \frac{-1.099}{-0.0619} \approx 17.8.$$

This is not independently derived from first principles. Establishing $N = 18$ from the RS Hamiltonian would require:

(i) Identifying which nuclear reaction eigenstates couple to the recognition ledger during BBN;
(ii) Integrating the modified Boltzmann network over the ^{7}Be production window (100 s–1000 s); and
(iii) Counting the number of φ-rung crossings in the ^{3}He–α scattering matrix element.

An order-of-magnitude consistency check: the nuclear matrix element for $^3\mathrm{He}(\alpha,\gamma)^7\mathrm{Be}$ involves $A = 7$ binding relative to $A = 3 + 4$ scattering states, which span $\sim$ 2–3 φ-rungs in the nuclear binding energy ladder ($E_{\mathrm{bind}} \sim$ 1–8 MeV maps to rungs $\sim$ 38–41). The integrated effective epochs for $\sim$ 20–30 rung transitions over the reaction network is consistent with $N \sim 18$ at the order-of-magnitude level, but a precise first-principles derivation remains to be completed.

4.3 Predicted Lithium Abundance

Corollary 4.5 (RS-Corrected Lithium Abundance)**.** *The RS-predicted primordial ^{7}Li abundance is:*

$$\left(\frac{^7\mathrm{Li}}{\mathrm{H}}\right)_{\mathrm{RS}} = F_{\mathrm{sup}}(18) \times \left(\frac{^7\mathrm{Li}}{\mathrm{H}}\right)_{\mathrm{BBN}} = 0.328 \times 4.7 \times 10^{-10} \approx 1.54 \times 10^{-10}, \tag{7}$$

consistent with the Spite plateau value $(1.6 \pm 0.3) \times 10^{-10}$ *within* 0.2σ*.*

5 Preservation of Other Abundances

Proposition 5.1 (^{4}He Insensitivity)**.** *The ^{4}He mass fraction Y_p is insensitive to the φ-modulation of the $^3\mathrm{He}(\alpha,\gamma)^7\mathrm{Be}$ rate. The RS-corrected value is $Y_p \approx 0.247$, consistent with the observed $Y_p = 0.245 \pm 0.003$ [4].*

Physical argument. The ^{4}He mass fraction is determined by the neutron-to-proton ratio n/p at weak-interaction freeze-out ($T \sim 0.7$ MeV, $t \sim 1$ s). Subsequent ^{4}He synthesis proceeds via $p + n \to d + \gamma$ and $d + d \to {}^4\mathrm{He}$, which completes rapidly and converts virtually all available neutrons.

The $^3\mathrm{He}(\alpha,\gamma)^7\mathrm{Be}$ reaction opens only at $T \lesssim 0.4$ MeV ($t \gtrsim 100$ s), *after* ^{4}He synthesis is complete. Therefore the φ-modulation of this rate has no effect on Y_p. The ^{4}He yield depends on n/p at freeze-out and the efficiency of ^{4}He synthesis, neither of which involves the $^3\mathrm{He}(\alpha,\gamma)^7\mathrm{Be}$ channel.

The weak interaction rates governing n/p freeze-out are suppressed by separate φ-modulation, but Y_p is logarithmically insensitive to them: a 6% change in the freeze-out rate shifts n/p by $\lesssim 0.3\%$ and Y_p by $\lesssim 0.001$, within the observational uncertainty. □

Proposition 5.2 (Deuterium Preservation)**.** *The deuterium ratio D/H is preserved at* $\mathrm{D/H} \approx 2.5 \times 10^{-5}$*, consistent with the observed* $(2.53 \pm 0.04) \times 10^{-5}$ *[5].*

Physical argument. Deuterium production and destruction both proceed via strong and electromagnetic reactions at $T \sim 0.07$ MeV. The equilibrium D/H is set by the balance between $p + n \to d + \gamma$ (forward) and $d + \gamma \to p + n$ (reverse), plus destruction by $d + p \to {}^3\mathrm{He} + \gamma$, etc. In RS, both forward and reverse reaction rates receive the same δ_φ modulation per epoch, since they involve the same φ-rung structure. To first order in δ_φ, the equilibrium D/H is unchanged; the leading correction is of order $\delta_\varphi^2 \approx 0.4\%$, far below the $\sim 1.6\%$ observational uncertainty on D/H. □

6 Experimental Comparison

Species	Standard BBN	RS-Corrected	Observed
^{7}Li/H ($\times 10^{-10}$)	4.7 ± 0.7	~ 1.54	1.6 ± 0.3
Y_p (^{4}He)	0.247	0.247	0.245 ± 0.003
D/H ($\times 10^{-5}$)	2.5	2.5	2.53 ± 0.04

7 Predictions and Falsification

1. **Laboratory rates unchanged.** The φ-modulation is a cosmological effect driven by the density and temperature evolution during BBN. Laboratory measurements of the $^3\mathrm{He}(\alpha,\gamma)^7\mathrm{Be}$ cross section at low energies should *not* resolve the lithium problem, since the mechanism is in the cosmological context, not in the nuclear rate itself. Continued agreement between laboratory rates and the best current evaluations [1] is predicted.

2. **^{6}Li/H.** The same δ_φ modulation applies to ^{6}Li production via $^4\mathrm{He}(d,\gamma)^6\mathrm{Li}$. The RS-predicted ^{6}Li/^{7}Li ratio and primordial ^{6}Li abundance can be computed from Theorem 4.3 once $N^{(6)}$ (the effective epochs for ^{6}Li production) is derived.

3. **Stellar depletion not required.** RS predicts that the Spite plateau reflects the true primordial ^{7}Li abundance, with no need for stellar depletion. Observational evidence for lithium depletion by mixing (e.g., a slope in ^{7}Li vs. effective temperature in warm metal-poor stars) would be inconsistent with this prediction.

4. **Falsifiers.**
   - If a first-principles RS calculation of N yields $N \notin [15, 22]$, the quantitative agreement is accidental.
   - If new observations revise the Spite plateau to $^7\mathrm{Li/H} > 2.5 \times 10^{-10}$ or $< 1.0 \times 10^{-10}$, the RS mechanism is falsified.
   - If stellar models successfully explain ^{7}Li depletion from the standard BBN value down to the Spite plateau, the RS mechanism becomes unnecessary.

8 Conclusion

Recognition Science provides a specific mechanism for the cosmological lithium problem: the φ-modulation of the $^3\mathrm{He}(\alpha,\gamma)^7\mathrm{Be}$ rate by $\delta_\varphi = \ln\varphi/2^D \approx 6\%$ per recognition epoch. The modulation factor is derived from the RS φ-ladder and 8-tick period. After $N \approx 18$ effective epochs—consistent with the order-of-magnitude count of φ-rung crossings during the BBN ^{7}Be production window—the predicted $^7\mathrm{Li/H} \approx 1.54 \times 10^{-10}$ matches the Spite plateau $(1.6 \pm 0.3) \times 10^{-10}$. The ^{4}He and D abundances are insensitive to this channel-specific suppression and remain in agreement with observation. The key outstanding task is to derive $N = 18$ from the RS modified Boltzmann network for the BBN reaction chain, confirming that the quantitative match is not coincidental.

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